

the recognition is what makes the structure close

THE ONE ASSUMPTION · A TIMELESS SELF-REFERENTIAL STRUCTURE

$$T(\mathcal{U}) = \mathcal{U}$$

description and described are one object · existence = self-consistency

THE FINITE CONDITIONS UNDER WHICH THE LOOP CLOSURES

AXIOM 1

twelve-port icosahedral
carrier net on the sphere,
complete reversible
port response

AXIOM 2

observers agree on
meaning under sharing,
translation, coarse-graining
 $Ad_U D(v) = D(a \cdot v)$

AXIOM 3

no structure beyond what
agreement forces
(information projection:
equal 1/12 port weights)

THE MACHINE (BOOK 1)

spherical topology $\rightarrow \Sigma(6 - \deg v) = 12 \rightarrow$ twelve ports $\rightarrow A_5$ of order 60
repair law: $L = 5I - A$, one round $T = I - L/60$, unique resting state
consensus protocol: read ports $\rightarrow$ compare $\rightarrow$ repair $\rightarrow$ record $\rightarrow$ checkpoint
public reality = what survives comparison and repair

CANDIDATE CLOSURES REQUIRE TYPED PHYSICAL SAME-CARRIER MAPS

LOCAL PIXEL FIXED POINT (conditional)

$$\mathcal{P} = \varphi + \sqrt{\pi} / A_T(\mathcal{P})$$

$$\mathcal{P}_\star = 1.630968209...$$

screen area $\leftrightarrow \alpha_{em}$: supplied same-carrier map
unique: interval Banach contraction 0.3041
unique inside declared map; physical bridge open

GLOBAL CAPACITY FIXED POINT (conditional)

$$N = \log M_0(\mathcal{U}_N)$$

$$\log(N/\pi) = 6\pi / [\mathcal{P} \cdot \alpha_U(\mathcal{P})] - \mathcal{P}/24$$

$$\text{candidate } N_\star = 3.3001 \times 10^{122}$$

horizon-record identification required
comparison coordinate; no cosmic output yet

$$\alpha_U(\mathcal{P})$$

$$\text{SI UNIT BRIDGE (not a closure): } \gamma_\star = \ell_\star \cdot v_{Cs}/c \approx 4.955974 \times 10^{-34}$$

$$\Rightarrow \ell_\star^2 = 2.612280 \times 10^{-70} \text{ m}^2 \cdot K_{\text{cell}} = 4N_\star/\mathcal{P}_\star \cdot \Lambda_\star \ell_\star^2 = 3\pi/N_\star$$

ROUTE A · CARRIER $\rightarrow$ SUPPORT

extract carrier and support readouts

1 · SOURCE POSET + HEIGHT

read-from order; canonical height (not clock)

2 · RANK-3 SOURCE QUOTIENT

unit directions = $S^2/\text{null rays}$
exact algebraic carrier rank three

3 · CONDITIONAL PLACEMENT

spatial readback + edge speed
converse support gives faithfulness

4 · PHYSICAL LIMIT INPUTS

count-volume + manifoldlikeness
topology + curvature convergence

SOURCE-DERIVED AMBIENT 1+3 CARRIER

$R \times V_\star$: ambient dim 4, $(+---)$; carrier independent of height
placement $e \mapsto (r \cdot \text{sourceHeight}(e), x(e))$; exact order $\mapsto$ cone
smooth manifold and physical causality: CONDITIONAL

FINITE EINSTEIN SHAPE

separate 3+1 tensor interface
9 balances + Ward/Bianchi

CONDITIONAL EFFECTIVE 3+1

signal causality + count-volume
refinement + manifold tests

CONDITIONAL SMOOTH GR PROMOTION

same-family curvature + stress; coupling, vacuum, physical scale

PHYSICAL LIMIT BOUNDARY

compatibility controls supply no source-selected manifold refinement

ROUTE B · PORT RESPONSE $\rightarrow A_5$ CURRENT

read the algebra off one local carrier

GAUGE-FORCING THEOREM

$$P_{12} = 1 \oplus 3 \oplus 3' \oplus 5, \text{ one fixed line, } 11 = 3 + 8$$

$$g = u(1) \oplus su(2) \oplus su(3)$$

$$1 + 3 + 8 = 12, \text{ the gauge adjoint}$$

GLOBAL FORM AND MATTER

$(SU(3) \times SU(2) \times U(1)) / \mathbb{Z}_6$ (adjoint form)
finite scan: 15 states, anomalies cancel, one Higgs
rank-three family band $\Rightarrow 15 \times 3 = 45$

$$\mathcal{P} \rightarrow \alpha_U \approx 0.04112 \rightarrow v = 246.767 \text{ GeV}$$

$$\alpha^{-1} = 137.035660 \text{ (} 2.5 \times 10^{-6} \text{)}$$

$$M_H = 125.1995 \text{ GeV}$$

$$\text{screen-grain reading: } \alpha^{-1} = 136.994835$$

chart (80.330, 91.119) $\rightarrow$ one-loop pole transport

$$W: 80.377000 \text{ Z: } 91.187978 \text{ GeV}$$

$$\text{top: } 172.352 \text{ GeV } \tau: 1776.969 \text{ MeV}$$

$$\text{face-sector Koide: } Q = 2/3 \text{ exactly}$$

SHARED BY BOTH ROUTES · QUANTUM RULES AND THE HISTORY LAW

agreement on records $\Rightarrow$ Born rule, Lüders conditioning, no-signaling
exact Tsirelson bound $2\sqrt{2}$ · central records shareable, others unclonable
one mean-action constraint $\Rightarrow p_\beta(\gamma) = q(\gamma) e^{-(\beta A(\gamma))}/Z_\beta$

THE LOOP CLOSURES · THE UNIVERSE RECOGNIZES ITSELF

physics $\rightarrow$ chemistry $\rightarrow$ life $\rightarrow$ minds $\rightarrow$ records $\rightarrow$ recovered architecture
observers rebuild the specification they were read off, and it emits the same records
 $\mathcal{P}_\star$ and $N_\star$ are unique candidates inside supplied maps

CONDITIONAL UNIFICATION TARGET

three axioms + closure equations + named physical bridge receipts
 $\alpha \cdot G \cdot c \cdot \Lambda \cdot M_H \cdot W \cdot Z \cdot \text{quarks} \cdot \text{leptons} \cdot \text{gauge group} \cdot \text{conditional 3+1 spacetime}$